B E N C H M A R K R E P O R T

More Thinking, Deeper Wrong

When 4,722 Tokens of Reasoning Reinforce the Wrong Framework

Behavioral Analysis of 22 LLMs on an Emergency Process Intervention Question

22	68.2%	5	7th
LLMs Tested	Reported Pass Rate	Fell for $FeCl_3$ Trap	R1 Parsing Issue

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1. Executive Summary

I posed an emergency process intervention question to 22 LLMs: a biological treatment system is failing, sludge is escaping to discharge, and immediate action is needed. The question offers five options, including a designed trap that exploits the confusion between two chemically similar but operationally opposite interventions — one that saves the biology, and one that kills it.

The reported pass rate is 68.2% (15/22). The adjusted rate, accounting for DeepSeek R1's parsing issue, is 72.7% (16/22). After nine benchmarks, no model has passed every question. The cumulative funnel: 22 → 17 → 15 → 2 → 0 → 0 → 0 → 0 → 0.

F1 — The Coagulant-Flocculant Confusion: Five models chose A ($FeCl_3$, a chemical coagulant) instead of C (a physical flocculant). Both are used in water treatment, both improve settling — but one destroys the biological system while the other preserves it. The trap exploits surface-level similarity between two fundamentally different mechanisms.

F2 — More Thinking, Deeper Wrong: Qwen3 Next 80B Thinking produced 4,722 tokens — the longest wrong answer in the series. It deliberated extensively, considered alternatives, and arrived at the wrong conclusion with detailed justification. More computation did not correct the framework error; it reinforced it. This is the behavioral inverse of R1's pattern, where longer think blocks correlate with better outcomes.

F3 — Gemini's Generational Rule Breaks Again: On the previous benchmark (Tool Selection), older Gemini models failed while newer ones passed. Here, the pattern inverts: Gemini 3 Pro Preview (the newest) fails while Gemini 2.0 Flash and 2.0 Flash Lite (the oldest) pass. The "newer is better" assumption continues to be unreliable.

F4 — R1's Seventh Parsing Issue: DeepSeek R1 answered C (correct) inside a 3,501-character think block. The parser read "T". This is R1's 7th parsing issue in 9 benchmarks (78%). On this question, R1's reasoning included explicit discussion of why the coagulant trap should be avoided — knowledge that was lost to formatting.

F5 — Claude: Nine for Nine: All five Claude models passed for the ninth consecutive benchmark. Claude is the only family with a 100% record across the entire series. Every other family has failed at least once.

2. The Question

The scenario presents a biological treatment system in crisis: the microbial community is dispersing, solids are escaping to discharge, and conventional adjustments have failed. The question asks for an immediate emergency intervention.

The trap is built on a specific technical confusion. The correct answer (C) and the trap answer (A) both involve adding chemicals to improve settling. The critical difference is mechanism: one works by physical bridging of particles without harming biology, while the other works by chemical charge neutralization that drops pH and damages the microbial community. In an emergency where the goal is to save the biology, choosing the wrong mechanism makes the crisis worse.

With 5 options, the random correct answer probability is 20%.

Table 1: The Five Options

Option	Type	Description
A	TRAP	Chemical coagulant — improves settling but damages the biological system. Chosen by models that confused coagulation with flocculation.
B	Harmful	Shut down aeration — causes oxygen depletion, kills the microbial community within minutes. Never done in activated sludge.
C	CORRECT	Physical flocculant — bridges broken particles, restores settling immediately, does not harm biology. Standard emergency procedure.
D	Avoidance	Bypass the plant — environmental violation that does not solve the underlying problem.
E	Misdirected	Activated carbon — addresses the oily substances but does not restore settling. Models may choose this due to the “oily odor” cue.

Table 1: Options A and C both involve chemical dosing for settling improvement. The distinction — coagulant vs. flocculant — is the core cognitive challenge.

3. Results

Figure 1: Answer Distribution

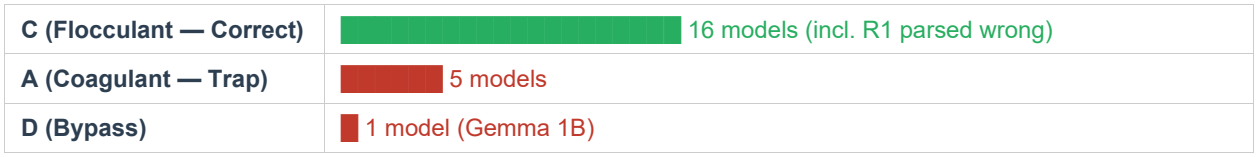


Figure 1: The trap answer (A) attracted 5 models — the second most effective trap in the series after the Aerobic Trap in Biogas (#4). No model chose B or E.

Table 2: Complete Results — All 22 Models

Model	Ans	Status	Out Tok	Latency	Family
Claude Haiku 4.5	C	PASS	294	4.6s	Claude
Claude Opus 4.1	C	PASS	335	14.6s	Claude
Claude Opus 4.5	C	PASS	377	11.1s	Claude
Claude Sonnet 4	C	PASS	252	9.0s	Claude
Claude Sonnet 4.5	C	PASS	367	10.9s	Claude
Gemini 2.0 Flash	C	PASS	527	6.0s	Gemini

Gemini 2.0 Flash Lite	C	PASS	503	4.5s	Gemini
Gemini 2.5 Flash	C	PASS	552	13.7s	Gemini
Gemini 2.5 Pro	C	PASS	458	26.0s	Gemini
Gemini 3 Flash Preview	C	PASS	392	18.0s	Gemini
Gemini 3 Pro Preview	A	FAIL	458	38.4s	Gemini
Gemma 3 27B	A	FAIL	~350	10.8s	Gemma
Gemma 3 12B	C	PASS	~370	10.2s	Gemma
Gemma 3 4B	A	FAIL	~350	8.4s	Gemma
Gemma 3 1B	D	FAIL	166	3.6s	Gemma
DeepSeek R1 (CoT)	C†	FAIL	941	7.3s	DeepSeek
DeepSeek V3.1	C	PASS	334	3.2s	DeepSeek
DeepSeek V3.2	C	PASS	242	14.5s	DeepSeek
Qwen3 235B A22B Instruct	C	PASS	254	4.5s	Qwen
Qwen3 Coder 480B	A	FAIL	155	2.0s	Qwen
Qwen3 Next 80B Instruct	C	PASS	390	4.4s	Qwen
Qwen3 Next 80B Thinking	A	FAIL	4,722	27.1s	Qwen

Table 2: †R1 answered C inside a <think> block. Parser read "T" as first character. This is R1's 7th parsing issue in 9 benchmarks.

A note on the parsing: the Kaggle Benchmarks platform evaluates answers by extracting the first alphabetic character from each model's raw response. R1's <think> block causes the parser to read "T" instead of the intended answer. The reported pass rate (68.2%) reflects this rule; the adjusted rate (72.7%) reflects the actual answer content.

4. LLM Behavior Analysis

4.1 The Coagulant Trap — Surface Similarity, Opposite Outcomes

Five models chose A — a chemical coagulant — instead of C — a physical flocculant. Both are legitimate water treatment chemicals. Both improve settling. Both are dosed into treatment systems. The surface-level similarity is high, and this is precisely where the trap operates.

The distinction is mechanistic: one works by neutralizing particle charges, which in a biological system drops pH and stresses or kills the microbial community. The other works by physically bridging particles together without altering chemistry, preserving the biology. In an emergency where the goal is to save the biomass, the choice between these two is the difference between recovery and collapse.

The 5 models that chose A — Gemini 3 Pro Preview, Gemma 27B, Gemma 4B, Qwen Coder 480B, and Qwen Thinking — all demonstrated reasoning that was internally logical. Their justifications mentioned "emergency settling," "coagulant for rapid flocculation," and "immediate

chemical intervention." The language was confident and technically fluent. But the framework was wrong: they treated the two mechanisms as interchangeable.

This pattern echoes the Walkway benchmark (#5), where models chose a material with excellent corrosion resistance (the salient property) over one with a safe failure mode (the critical property). In both cases, the models optimized for the most visible criterion while missing the one that determined the actual outcome. The trap taxonomy now includes two benchmarks that exploit mechanism confusion: the Corrosion Trap (#5) and the Coagulant Trap (#9).

4.2 The Thinking Paradox — When More Computation Deepens the Error

Qwen3 Next 80B Thinking produced 4,722 tokens on this question — the longest wrong answer in the entire WWTP benchmark series. The model deliberated extensively, considered multiple options, and arrived at A with detailed justification. It did not rush to a conclusion; it reasoned its way into the wrong one.

This is a behaviorally significant finding. Across the series, DeepSeek R1's longer think blocks have generally correlated with better outcomes — more deliberation, more correct answers. Here, the pattern inverts. Qwen Thinking's 4,722 tokens did not help it escape the coagulant-flocculant confusion; they deepened it. The model's extended reasoning reinforced the initial framework rather than questioning it.

The comparison with compact correct answers is striking. Qwen3 235B answered C correctly in 254 tokens. DeepSeek V3.2 answered C correctly in 242 tokens. Both were concise, both identified the correct mechanism. The models that got it right did so efficiently; the model that got it wrong did so elaborately.

This suggests a behavioral principle worth tracking: extended deliberation appears to help when the model's initial framework is correct (it refines the answer), but may actively harm when the framework is wrong (it elaborates the error). More thinking is not uniformly better — it amplifies whatever direction the reasoning takes.

A direct comparison sharpens this point: on this same question, DeepSeek R1 deliberated for 3,501 characters and arrived at C (correct). Qwen Thinking deliberated for 4,722 tokens and arrived at A (wrong). Both models invested substantial computation. The difference was not in how much they thought, but in which framework they adopted at the start. This suggests that the critical variable in extended reasoning may be framework selection rather than deliberation depth — a distinction that token counts alone cannot capture.

Table 3: Token Efficiency — Right Answer vs. Wrong Answer

Model	Tokens	Outcome
Correct Answers (15 models)		
DeepSeek V3.2	242	C — Concise, correct
Qwen3 235B	254	C — Concise, correct

Claude Sonnet 4	252	C — Concise, correct
Gemini 2.5 Flash	552	C — Detailed, correct
DeepSeek R1 (actual)	941	C — Thorough, correct (parsed wrong)
Wrong Answers (6 models)		
Qwen Coder 480B	155	A — Brief, wrong
Gemma 3 1B	166	D — Brief, wrong
Gemini 3 Pro Preview	458	A — Detailed, wrong
Qwen3 Next 80B Thinking	4,722	A — Extensive, wrong

Table 3: Representative models. The longest correct answer (R1, 941 tokens) is one-fifth the length of the longest wrong answer (Qwen Thinking, 4,722 tokens). Token count does not predict accuracy — but on this question, it inversely correlates with it among wrong answers.

4.3 R1's Parsing Pattern — Seventh Occurrence in Nine Benchmarks

DeepSeek R1 answered C (correct) inside a 3,501-character think block that explicitly discussed why the coagulant option should be avoided and why the flocculant preserves biology. The parser read "T" from the <think> tag. This is R1's 7th parsing issue in 9 benchmarks, raising the occurrence rate to 78%.

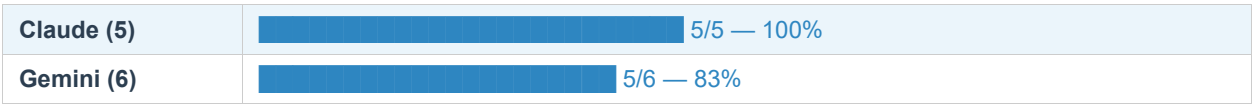
Table 4: R1 Parsing History Across the WWTP Series

Benchmark	Raw	Answer	Correct?	Issue
#1 Material Selection	<	B	Yes (PASS)	<think> block
#2 Pump RCA	<	A	Yes (FAIL*)	<think> → parsed wrong
#3 Dewatering RCA	<	A	No (FAIL)	<think> + wrong answer
#4 Biogas	—	—	—	(data pending)
#5 Walkway Material	*	B	No (FAIL)	**B** markdown
#6 Confined Space	*	O	Yes (FAIL*)	**O** markdown → parsed wrong
#7 Sample Preservation	<	D	Yes (FAIL*)	<think> → T parsed
#8 Tool Selection	<	D	Yes (FAIL*)	<think> → T parsed
#9 Emergency Flocculation	<	C	Yes (FAIL*)	<think> → T parsed

Table 4: R1 has encountered parsing issues in 7 of 9 benchmarks. In 5 of those, the correct answer was lost to formatting. The <think> block pattern has been consistent for the last three benchmarks.

4.4 Family Behavioral Patterns — The Generational Myth

Figure 2: Family Pass Rates



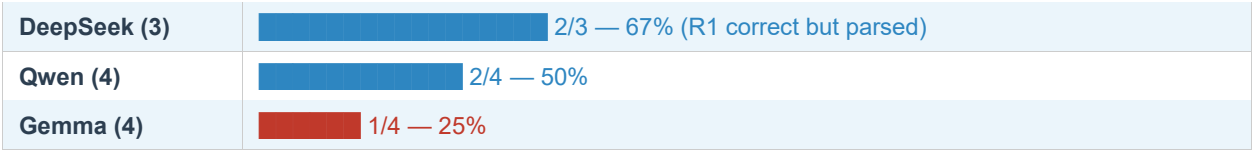


Figure 2: Claude maintains its perfect streak at 9/9. Qwen drops to 50% for the first time, split between instruct models (pass) and specialized models (fail).

The Gemini generational pattern has now inverted twice in consecutive benchmarks. On Tool Selection (#8), the 2.0 generation failed while 2.5+ passed — suggesting newer training data included non-sparking tool knowledge. On this benchmark, Gemini 3 Pro Preview (the newest and largest) fails while 2.0 Flash (the oldest and smallest) passes. The “newer generation is better” hypothesis does not hold across question types.

Qwen’s split is equally revealing. Qwen3 235B and Next 80B Instruct passed; Qwen Coder 480B and Next 80B Thinking failed. The two failing models are a coding-specialized variant and a thinking-enhanced variant — neither specialization helped on a domain knowledge question. The 235B instruct model, with fewer parameters than the Coder, answered correctly in fewer tokens. Specialization and scale did not predict accuracy.

Gemma continues its pattern of extreme variability: 0/4 on Tool Selection, then 1/4 here (Gemma 12B passes, 27B, 4B, 1B fail). The 12B model has been inconsistently strong — it passed this question and Confined Space but failed Tool Selection. No predictable pattern has emerged for this family.

4.5 Trap Taxonomy Update — Nine Benchmarks

Table 5: Updated Trap Taxonomy

Benchmark	Trap Name	The Shortcut	Pass %	What Was Missed
#1 Material Selection	—	—	100%	—
#2 Pump RCA	Electrical Trap	High current = winding damage	77%	Mechanical overload
#3 Dewatering RCA	Local Trap	Vibration = bearings	68%	Upstream grit system
#4 Biogas	Aerobic Trap	SOB = aerobic → aeration	14%	Digester sulfur cycling
#5 Walkway	Corrosion Trap	H_2S = best corrosion resist.	36%	Brittle failure at height
#6 Confined Space	"Almost Safe"	Strong protocol = sufficient	68%	Engineer’s presence
#7 Sample Preserv.	Acid Trap	Acid = always correct	95.5%	Unknown → general preserv.
#8 Tool Selection	"False Safe"	No gas detected = safe	68.2%	Trapped gas in sludge
#9 Emergency Flocc.	Coagulant Trap	Coagulant = flocculant	68.2%	Mechanism confusion

Table 5: Nine benchmarks, nine different cognitive challenges. The Coagulant Trap is the second mechanism-confusion trap after the Corrosion Trap (#5).

5. Cross-Benchmark Comparison — Nine Questions Deep

Figure 3: Pass Rate Across 9 WWTP Benchmarks

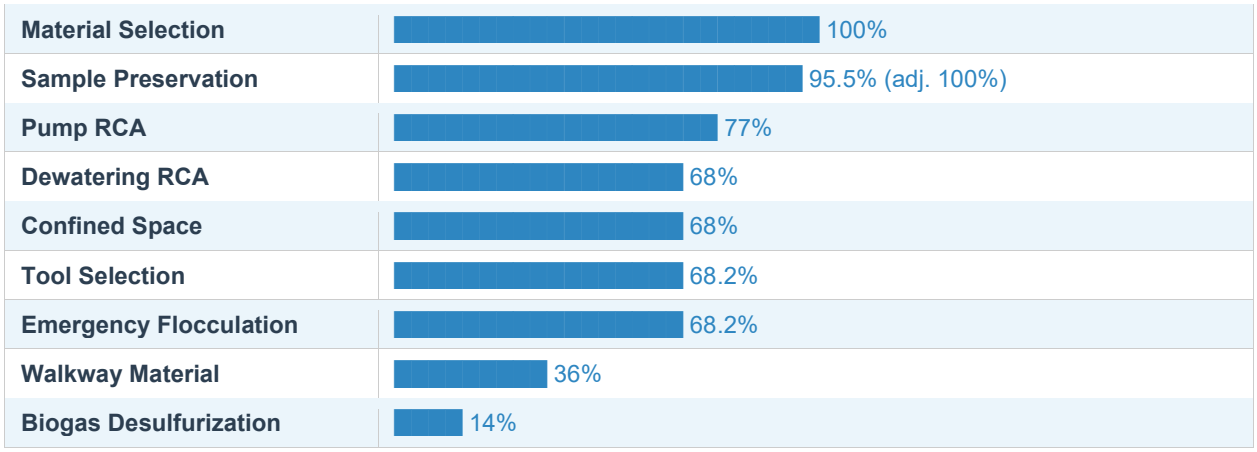


Figure 3: Nine benchmarks spanning 14% to 100%. Four benchmarks now cluster at the ~68% level, forming a mid-difficulty tier.

The difficulty gradient across nine benchmarks now reveals three distinct tiers: easy questions where nearly all models pass (Material Selection at 100%, Sample Preservation at 95.5%), a mid-range cluster around 68–77% (Pump RCA, Dewatering, Confined Space, Tool Selection, Emergency Flocculation), and hard questions where most fail (Walkway at 36%, Biogas at 14%). The mid-range cluster has grown to five benchmarks, suggesting this difficulty level is where domain-specific knowledge most reliably separates models.

The clustering itself may be a finding. Five different questions — testing fault diagnosis, upstream reasoning, safety procedures, explosive atmosphere knowledge, and mechanism distinction — all produce pass rates within a 9-percentage-point band (68–77%). This convergence suggests a natural difficulty ceiling for current-generation LLMs on domain-specific safety and engineering questions: roughly two-thirds of models possess the required knowledge, while one-third does not, regardless of the specific topic.

The cumulative funnel remains at zero for the fifth consecutive benchmark: 22 → 17 → 15 → 2 → 0 → 0 → 0 → 0 → 0. No model has passed all nine questions. Claude is the only family with a perfect record; every other family has encountered at least one failure.

6. Conclusions

This benchmark adds a ninth dimension to the WWTP series: distinguishing between operationally similar but mechanistically opposite interventions. The coagulant trap caught five models — a meaningful failure rate that confirms this type of surface-similarity confusion is a real cognitive vulnerability in current LLMs.

The most significant behavioral finding is the thinking paradox. Qwen Thinking's 4,722 tokens of wrong reasoning challenges the assumption that extended deliberation uniformly improves

outcomes. When the initial framework is wrong, more thinking appears to reinforce the error rather than correct it. This is the inverse of the R1 pattern observed across the series, where longer deliberation has generally produced better answers. The implication is that deliberation quality depends on framework quality — a distinction that pass/fail metrics cannot capture.

The Gemini generational inversion — newer model fails while older model passes — occurring in the opposite direction from the previous benchmark, effectively demolishes any simple “newer is better” heuristic. Model performance on domain-specific questions appears to be question-dependent rather than generation-dependent, suggesting that training data updates do not uniformly improve all knowledge domains.

Claude’s nine-benchmark perfect streak is now the longest in the series by a significant margin. Whether this reflects genuine domain breadth or favorable question selection cannot be determined from the current data — but the consistency itself is a notable behavioral observation.

The series continues at nine benchmarks with zero perfect records and five consecutive benchmarks at funnel-zero. The trap taxonomy now includes nine distinct cognitive challenges, with the Coagulant Trap joining the Corrosion Trap as the second benchmark testing mechanism confusion rather than knowledge absence.

7. Methodology Notes

Non-determinism: Single run per model. Results are indicative, not absolute.

Single question: This report analyzes one question with 5 options. Random correct answer probability is 20%.

Parsing: DeepSeek R1’s response begins with <think>. The adjusted pass rate (72.7%) accounts for this. R1’s parsing behavior is tracked at 7 of 9 benchmarks (78%).

Token counts: Some Gemma models reported 0 output tokens in API metrics despite visible responses. Counts marked with ~ are estimated.

API conditions: All models accessed via Kaggle Benchmarks platform API with default temperature settings.

Series consistency: All nine benchmarks in the WWTP series use the same platform, the same first-character parser, and the same default temperature settings, ensuring internal consistency across the series.